

Peishi Jiang

Assistant Professor · The University of Alabama

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EMPLOYMENT & APPOINTMENTS

- **Assistant Professor** 2025 - present
The University of Alabama, CCEE
- **Research Scientist (Levels 2 and 3)** 2021 - 2025
Pacific Northwest National Laboratory
- **Machine Learning Researcher** 2021
NASA Frontier Development Laboratory
- **Post Doctorate Research Associate (Level 1)** 2019 - 2021
Pacific Northwest National Laboratory

EDUCATION

- **University of Illinois at Urbana-Champaign** Urbana, IL
Ph.D. in Civil Engineering 2014 - 2019
Advisor: Professor Praveen Kumar
Dissertation title: Causal History Analysis of Complex System Dynamics. [[Link](#)]
- **Hong Kong University of Science & Technology** Hong Kong
M.Phil. in Civil Engineering 2012 - 2013
Advisor: Professor Yeou-Koung Tung
Thesis title: Establishment of Rainfall Depth-Duration-Frequency Relationships at Conventional Raingauges & Ungauged Sites in Hong Kong. [[Link](#)]
- **Zhejiang University** Hangzhou, China
B.E. in Water Resources and Ocean Engineering 2007 - 2011

RESEARCH INTERESTS

- Hydrology/Ecohydrology
- Causal inference using information theory
- Spatio-temporal modeling using deep learning
- Differentiable modeling in geosciences
- Watershed hydro-biogeochemistry modeling
- Sensitivity analysis and data assimilation
- Frequency analysis of hydrological extremes
- Hydrosystems engineering uncertainty analysis

PEER-REVIEWED PUBLICATIONS

1. [WRR25] **Peishi Jiang**, Patrick Kidger, Toshiyuki Bandai, Dennis Baldocchi et al. (2025), “JAX-CanVeg: A Differentiable Land Surface Model”, *Water Resources Research*, doi.org/10.1029/2024WR038116.
2. [JH24] **Peishi Jiang**, Pin Shuai, Alexander Y. Sun, and Xingyuan Chen (2024), “Optimizing Parameter Learning and Calibration in an Integrated Hydrological Model: Impact of Observation Length and Information”, *Journal of Hydrology*, doi.org/10.1016/j.jhydrol.2024.131889.
3. [JAMES23] **Peishi Jiang**, Zhao Yang, Jiali Wang, Chenfu Huang, Pengfei Xue, TC Chakraborty, Xingyuan Chen, and Yun Qian (2023), “Efficient Super-Resolution of Near-Surface Climate Modeling Using the Fourier Neural Operator”, *Journal of Advances in Modeling Earth Systems*, doi.org/10.1029/2023MS003800.

4. [HESS23] **Peishi Jiang**, Pin Shuai, Alexandar Sun, Maruti K Mudunuru, and Xingyuan Chen (2023), “Knowledge-Informed Deep Learning for Hydrological Model Calibration: An Application to Coal Creek Watershed in Colorado”, *Hydrology and Earth System Sciences*, doi.org/10.5194/hess-27-2621-2023.
5. [FES22] **Peishi Jiang**, Xingyuan Chen, Justine EC. Missik, Brittany A Verbeke, Heping Liu, and Zhongming Gao (2022), “Encoding diel hysteresis and the Birch effect in dryland soil respiration models through knowledge-guided deep learning”, *Frontiers in Environmental Science*, doi.org/10.3389/fenvs.2022.1035540.
6. [WRR22] **Peishi Jiang**, Kyongho Son, Maruti K. Mudunuru, and Xingyuan Chen (2022), “Using Mutual Information for Global Sensitivity Analysis on Watershed Modeling”, *Water Resources Research*, doi.org/10.1029/2022WR032932.
7. [EMS21] **Peishi Jiang**, Xingyuan Chen, Kewei Chen, Jeffrey Anderson, Nancy Collins, and Mohamad EL Gharamti (2021) “DART-PFLOTRAN: An Ensemble-based Data Assimilation System for Estimating Subsurface Flow and Transport Model Parameters”, *Environmental Modelling and Software*, doi.org/10.1016/j.envsoft.2021.105074.
8. [NeurIPS21] **Peishi Jiang**, Nis Meinert, Helga Jordão, Constantin Weisser, Simon Holgate, Alexander Lavin, Björn Lütjens, Dava Newman, Haruko Wainwright, Catherine Walker, Patrick Barnard (2021) “Digital Twin Earth–Coasts: Developing a fast and accurate physics-informed surrogate model for coastal floods via neural operators”, *2021 NeurIPS Workshop on Machine Learning for the Physical Sciences (ML4PS)*, arXiv:2110.07100.
9. [Entropy20] **Peishi Jiang** and Praveen Kumar (2020) “Bundled Causal History Interaction”, *Entropy*, doi.org/10.3390/e22030360.
10. [WRR20] **Peishi Jiang** and Praveen Kumar (2020) “Using Information Flow for Whole System Understanding from Component Dynamics”, *Water Resources Research*, doi.org/10.1029/2019WR025820.
11. [PRE19] **Peishi Jiang** and Praveen Kumar (2019) “Information Transfer from Causal History in Complex System Dynamics”, *Physical Review E*, doi:10.1103/PhysRevE.99.012306.
12. [PRE18] **Peishi Jiang** and Praveen Kumar (2018) “Interactions of Information Transfer Along Separable Causal Paths”, *Physical Review E*, doi:10.1103/PhysRevE.97.042310.
13. [EMS17] **Peishi Jiang**, Mostafa Elag, Praveen Kumar, Scott Dale Peckham, Luigi Marini, and Liu Rui (2017) “A Service-oriented Architecture for Coupling Web Service Models Using the Basic Model Interface (BMI)”, *Environmental Modelling & Software*, doi:10.1016/j.envsoft.2017.01.021.
14. [SERRA15] **Peishi Jiang** and Yeou-Koung Tung (2015) “Incorporating Daily Rainfalls to Derive Rainfall DDF Relationships at Ungaged Sites in Hong Kong and Quantifying Their Uncertainty”, *Stochastic Environmental Research and Risk Assessment*, doi:10.1007/s00477-014-0915-2.
15. [JH13] **Peishi Jiang** and Yeou-Koung Tung (2013) “Establishing Rainfall Depth-Duration-Frequency Relationships at Daily Raingauge Stations in Hong Kong”, *Journal of Hydrology*, doi:10.1016/j.jhydrol.2013.09.037.
16. [GMD24] Katherine Muller, **Peishi Jiang** et al. (2024), “Lambda-PFLOTRAN: Workflow for Incorporating Organic Matter Chemistry Informed by Ultra High Resolution Mass Spectrometry into Biogeochemical Modeling”, *Geoscientific Model Development*, doi.org/10.5194/gmd-2024-34.

17. [WRR24Sun] Alexander Y Sun, **Peishi Jiang**, Pin Shuai, and Xingyuan Chen (2024) “Bridging Hydrological Ensemble Simulation and Learning Using Deep Neural Operators”, *Water Resources Research*, doi.org/10.1029/2024WR037555.
18. [GRL24] Alexander Y. Sun, Himanshu Save, Ashraf Rateb, **Peishi Jiang**, and Bridget R. Scanlon. (2024), “Deciphering the Role of Total Water Storage Anomalies in Mediating Regional Flooding”, *Geophysical Research Letters*, doi:10.1029/2023GL108126.
19. [WRR24Bandai] Toshiyuki Bandai, Teamrat A. Ghezzehei, **Peishi Jiang**, et al. (2024), “Integrated Effects of Site Hydrology and Vegetation on Exchange Fluxes and Nutrient Cycling at a Coastal Terrestrial-Aquatic Interface”, *Water Resources Research*, doi.org/10.1029/2024WR037318.
20. [WRR24Bing] Bing Li, Zhi Li, Jianqiu Zheng, **Peishi Jiang**, et al. (2024), “Integrated Effects of Site Hydrology and Vegetation on Exchange Fluxes and Nutrient Cycling at a Coastal Terrestrial-Aquatic Interface”, *Water Resources Research*, doi.org/10.1029/2023WR035580.
21. [JVSTA24] Yadong Zhou, **Peishi Jiang**, et al. (2024), “Novel principal component analysis tool based on PYTHON for analysis of complex spectra of time-of-flight secondary ion mass spectrometry”, *Journal of Vacuum Science & Technology A*, doi.org/10.1116/6.0003355.
22. [HP23] Pin Shuai, **Peishi Jiang**, Ethan Coon, and Xingyuan Chen (2023), “The Importance of Explicitly Representing the Streambed in Watershed Models”, *Hydrological Processes*, doi.org/10.1002/hyp.15043.
23. [JH23] Moses B. Adebayo, Vanessa L. Bailey, Xingyuan Chen, Anya M. Hopple, **Peishi Jiang**, et al. (2023), “A hydrogeophysical framework to assess infiltration during a simulated ecosystem-scale flooding experiment”, *Journal of Hydrology*, doi.org/10.1016/j.jhydro1.2023.130243.
24. [FES22Mudunuru] Maruti K. Mudunuru, Kyongho Son, **Peishi Jiang**, Glenn Hammond, and Xingyuan Chen (2022), “Scalable deep learning for watershed model calibration”, *Frontiers in Earth Science*, doi.org/10.3389/feart.2022.1026479.
25. [HESS22] Alexander Y Sun, **Peishi Jiang**, Zong-Liang Yang, Yangxinyu Xie, and Xingyuan Chen (2022) “A graph neural network (GNN) approach to basin-scale river network learning: the role of physics-based connectivity and data fusion”, *Hydrology and Earth System Sciences*, doi.org/10.5194/hess-26-5163-2022.
26. [WRR22Chen] Kewei Chen, Xingyuan Chen, Xuehang Song, Martin A. Briggs, **Peishi Jiang**, Pin Shuai, Glenn Hammond, Hongbin Zhan, and John M. Zachara (2022), “Using Ensemble Data Assimilation to Estimate Transient Hydrologic Exchange Flow Under Highly Dynamic Flow Conditions”, *Water Resources Research*, doi.org/10.1029/2021WR030735.
27. [WRR21] Alexander Y Sun, **Peishi Jiang**, Maruti K Mudunuru, Xingyuan Chen (2021) “Explore Spatio-Temporal Learning of Large Sample Hydrology Using Graph Neural Networks”, *Water Resources Research*, doi.org/10.1029/2021WR030394.
28. [FES21] Erol LD Cromwell, Pin Shuai, **Peishi Jiang**, Ethan Coon, Scott L Painter, David Moulton, Youzuo Lin, and Xingyuan Chen (2021) “Estimating Watershed Subsurface Permeability From Stream Discharge Data Using Deep Neural Networks”, *Frontiers in Earth Science*, doi.org/10.3389/feart.2021.613011.
29. [WRR20Goodwell] Allison Goodwell, **Peishi Jiang**, Ben Ruddel and Praveen Kumar (2020) “DOEs Information Theory Provide a New Paradigm for Earth Science? Understanding Causality and Interaction.”, *Water Resources Research*, doi.org/10.1029/2019WR024940.

CONFERENCE PROCEEDINGS AND WHITE PAPERS

1. Sivasankaran Rajamanickam, Prasanna Balaprakash, **Peishi Jiang**, Jiali Wang and Nathan Urban (2022) “16 Hybrid Modeling”, Artificial Intelligence for Earth System Predictability (AI4ESP) Workshop Report, No. ANL-22/54. Argonne National Lab.(ANL), Argonne, IL, <https://doi.org/10.2172/1888810>.
2. **Peishi Jiang**, Xingyuan Chen, Maruti Mudunuru, Praveen Kumar, Pin Shuai, Kyongho Son and Alexander Sun (2021) “Towards Trustworthy and Interpretable Deep Learning-assisted Ecohydrological Models”, DOE Artificial Intelligence for Earth System Predictability workshop, <https://doi.org/10.2172/1769787>.
3. **Peishi Jiang** and Yeou-Koung Tung (2013) “Quantification of rainfall DDF relationships at ungauged sites in Hong Kong”, World Environmental and Water Resources Congress 2013 Showcasing the Future, ASCE, doi:10.1061/9780784412947.323.

TEACHING EXPERIENCES

- **CLIMATE405, ML in Earth & Environmental Sciences** University of Michigan
Guest Lecturer 2024
- **PNNL ASGC AI/ML Bootcamp** PNNL
Instructor 2022
- **CEE202 Engineering Risk and Uncertainty** UIUC
Teaching Assistant 2018
- **CEE598NLH Nonlinear Methods in Hydrology** UIUC
Teaching Assistant 2017
- **CIVL3520 Hydraulics** HKUST
Teaching Assistant 2013
- **CIVL3530 Hydrology** HKUST
Teaching Assistant 2012

MENTORING EXPERIENCES

- **Doctoral Thesis Committee** University of Toledo
Member 2024 - present
- **Mentoring Division New Hires** PNNL
Peer Partner 2023 - present
- **PNNL High School Intern Program** PNNL
Mentor 2020

GRANT & PROPOSAL WRITING

- [Funded] **River Corridor Science Focus Area Renewal** DOE
Cross-Cut Lead (Total: \$20,000,000; PI: James Stegen) 2024 - 2028
Project: River Corridor Hydro-Biogeochemistry from Molecular to Multi-Basin Scales
- [Funded] **PNNL Seed Laboratory Directed R&D** DOE
PI (Total: \$110,000) 2023 - 2025
Project: A Differentiable Hybrid Land Model for Seamlessly Integrating Deep Learning with Physics-based Components
- [Funded] **PNNL Generative AI Laboratory Directed R&D** DOE
Co-PI (Total: \$50,000; PI: Lingcheng Li) 2024

Project: Innovative Kilometer-scale Climate Downscaling using Generative AI

- [Unfunded] **PNNL Seed Laboratory Directed R&D** DOE
PI (Total: \$150,000) 2024 - 2025
Project: Understanding Ecohydrological Responses to Warming Climate Using a Multiscale Hybrid Terrestrial Ecohydrological Model
- [Unfunded] **Scientific Discovery through Advanced Computing (SciDAC)** DOE
Task Lead (PI: Auroop R. Ganguly) 2022
Project: Propagation of Uncertainty and Causality at Ocean-Land-Atmosphere Model Interfaces (Prop-OLA)
- [Unfunded] **Argonne Maria Goeppert Mayer Fellowship** DOE
PI (Declined at the finalist) 2019
Project: A Domain-Informed Scientific Machine Learning Approach for Environmental Systems Using Information Theory

OTHER RESEARCH PROJECTS INVOLVED

- **River Corridor Science Focus Area** DOE
Researcher, Task Lead (PI: Tim Scheibe) 2020 - 2024
Project: River Corridor Hydro-Biogeochemistry from Reaction to Basin Scale
- **ExaSheds** DOE
Researcher (PI: Carl Steefel) 2020 - 2024
Project: Advancing Watershed System Understanding through Exascale Simulation and Machine Learning
- **PNNL IDEAS-Watersheds** DOE
Researcher (PI: Xingyuan Chen) 2019 - Present
Project: The Interoperable Design of Extreme-scale Application Software Watersheds
- **IML-CZO** NSF
Graduate Student Researcher (PI: Praveen Kumar) 2015 - 2019
Project: Intensively Managed Landscapes Critical Zone Observatory
- **EarthCube Geo-Semantics** NSF
Graduate Student Researcher (PI: Praveen Kumar) 2014 - 2015
Project: A Geo-Semantic Framework for Integrating Long-Tail Data and Models

PROFESSIONAL AND COMMUNITY SERVICES

- **Associate Editor** 2025
Journal of Hydrology
- **US Department of Energy (DOE) Proposal Reviewer** 2024
DOE Office of Science, Biological and Environmental Research (BER) Program
- **PNNL Proposal Reviewer** 2022
DOE Office of Science, BER Early Career Research Program
- **Conference Organizer** 2024
The 3rd HydroML Symposium at PNNL
- **Hackathon Organizer** 2024
The PNNL ASGC AI/ML Hackathon

- **Primary Session Convener** 2024
2024 AGU Fall Meeting, Causality in the Geosciences
- **Session Convener** 2022, 2023
2022, 2023 AGU Fall Meetings, Causal Inference in Hydrology
- **Session Chair** 2021
DOE AI4ESP Workshop, Hybrid Modeling
- **Committee Member** 2017 - 2018
AGU Hydrological Uncertainty Technical Committee
- **Journal Reviewer:** Nature Scientific Reports, npj Climate and Atmospheric Science, Water Resources Research, Hydrology and Earth System Sciences, Journal of Hydrology, Journal of Hydrometeorology, Agricultural and Forest Meteorology, Journal of Climate, Weather and Forecasting, Journal of Machine Learning for Modeling and Computing, Advances in Water Resources, Computers Environment and Urban Systems, Journal of Hydroinformatics, Journal of Geophysical Research: Atmospheres, International Journal of Digital Earth, Artificial Intelligence for the Earth Systems, Frontiers in Water, and Stochastic Environmental Research and Risk Assessment

PRESENTATIONS

1. [Invited] Coastal Sustainability Workshop at the University of Houston 2025, *Unraveling Water Stress Impacts on Canopy Water and Carbon Fluxes via a Hybrid Land Model*, Houston, TX.
2. [Invited] CCEE Seminar at the University of Alabama 2025, *New Paradigms in Watershed Science: Embracing the Data-Driven Era*, Tuscaloosa, AL.
3. [Invited] CE Seminar at Clemson University 2025, *New Paradigms in Watershed Science: Embracing the Data-Driven Era*, Clemson, SC.
4. [Invited] CEE Seminar at the University of Missouri 2024, *New Paradigms in Watershed Science: Embracing the Data-Driven Era*, Columbia, MO.
5. [Invited] PNNL EBSD Directorate Advisory Committee Meeting 2024, *Shark Tank Presentations: Wildfire Impacts on Terrestrial Water and Carbon Dynamics*, Pacific Northwest National Laboratory, WA.
6. [Invited] HydroML Symposium 2024, *JAX-CanVeg: A Differentiable Land Surface Model*, Richland, WA.
7. [Invited] ASGC Division Seminar 2022, *Applications of Deep Learning in Watershed Modeling*, Pacific Northwest National Laboratory, WA.
8. [Invited] USGS Integrated Modeling Team Seminar 2022, *Neversink Watershed Modeling*, Online.
9. [Invited] Environmental Science Presentation 2018, *An Information-Theoretic Framework for Unraveling the Dynamics of Self-Organization in Ecosystem*, Argonne National Laboratory, IL.
10. [Oral] AGU Fall Meeting 2024, *Small Tool, Big Dream: A Knowledge-Informed Mapping (KIM) Toolkit to Power the Model-Experiment Cycle*, Washington, DC.
11. [Poster] DOE RC-SFA Review 2024, *ST-3A: River Corridor Model (RCM) Development and ERsed Allometry Dynamics in the YRB*, Rockville, MD.
12. [Oral] DOE ESS PI Meeting 2024, *Potential of Using Remote Sensing-based Foundation Models in Addressing Environmental Problems*, Reston, VA.

13. [Poster] DOE ESS PI Meeting 2024, *Integrating Organic Matter Measurements into Watershed Hydro-Biogeochemical Models*, Reston, VA.
14. [Oral] AGU Fall Meeting 2023, *Differentiable Land Surface Modeling Using JAX*, San Francisco, CA.
15. [Poster] DOE ESS PI Meeting 2023, *Integrated Modeling of Carbon and Nitrogen Cycling at the Yakima River Basin*, Bethesda, MD.
16. [Poster] AGU Fall Meeting 2022, *Zero-Shot Super-Resolution of WRF Surface Heat Simulation Using Fourier Neural Operator*, Chicago, IL.
17. [Poster] HydroML Symposium 2022, *Hydrological Model Calibration Using Knowledge-Informed Deep Learning at Coal Creek Watershed, CO*, Penn State University, State College, PA.
18. [Oral] AGU Fall Meeting 2021, *Building up Trustworthiness of Deep Learning-based Emulator for Environmental Process-based Model*, Online.
19. [Poster] 2021 NeurIPS Workshop on Machine Learning for the Physical Sciences, *Digital Twin Earth-Coasts: Developing a fast and physics-informed surrogate model for coastal floods via neural operators*, Online.
20. [Oral] PNNL Physics Informed Machine Learning Workshop, *Leveraging the diel hysteresis between soil respiration and temperature to improve Rsoil modeling using bidirectional LSTM*, Online.
21. [Oral] AGU Fall Meeting 2020, *DART-PFLOTTRAN: An Ensemble Data Assimilation System for Modeling Subsurface Process*, Online.
22. [Oral] AGU Fall Meeting 2019, *Using Information Flow for Whole System Understanding from Component Dynamics*, San Francisco, CA.
23. [Poster] AGU Fall Meeting 2018, *Unraveling the Causal Dynamics in Stream Solutes by Using an Information-Theoretic Approach*, Washington, D.C..
24. [Oral] Entropy 2018: From Physics to Information Sciences and Geometry, *Interactions of Information Transfer Along Separable Causal Paths*, Barcelona, Spain.
25. [Poster] AGU Fall Meeting 2017, *Understanding Information Flow Interaction along Separable Causal Paths in Environmental Signals*, New Orleans, CA.
26. [Poster] CSDMS Annual Meeting 2017, *A Service-Oriented Architecture for Coupling Web Service Models Using the Basic Model Interfaces*, Boulder, CO.
27. [Oral] AGU Fall Meeting 2016, *An Application of the Geo-Semantic Micro-services in Seamless Data-Model Integration*, San Francisco, CA.
28. [Poster] AGU Fall Meeting 2015, *A Smart Modeling Framework for Integrating BMI-enabled Models as Web Services*, San Francisco, CA.
29. [Poster] CSDMS Annual Meeting 2015, *GeoSemantic Resource Alignment Service*, Boulder, CO.
30. [Oral] World Environmental and Water Resources Congress (EWRI 2013), *Quantification of Rainfall DDF Relationships at Ungauged Sites in Hong Kong*, Cincinnati, OH.
31. [Oral] 2012 IAHR-HK Student Research Forum, *Establishing Rainfall Depth-Duration-Frequency Relationships at Daily Raingauge Stations in Hong Kong*, Hong Kong.

OPEN-SOURCE SOFTWARES & DATASETS

1. [Github] **Peishi Jiang** (Sept 2024 online) *JAX-CanVeg: A Differentiable Land Surface Model*. github.com/pnnl/JAX-CanVeg.
2. [Zenodo] **Peishi Jiang** et al. (June 2024 online) *Data and scripts associated with “Optimizing Parameter Learning and Calibration in an Integrated Hydrological Model: Impact of Observation Length and Information”*. doi.org/10.5281/zenodo.12213081.
3. [Zenodo] **Peishi Jiang** et al. (July 2023 online) *Data and scripts associated with “Knowledge-informed deep learning for hydrological model calibration: an application to Coal Creek Watershed in Colorado”*. doi.org/10.5281/zenodo.8128089.
4. [Zenodo] **Peishi Jiang** et al. (May 2023 online) *Scripts associated with the manuscript “Efficient Super-Resolution of Near-Surface Climate Modeling Using the Fourier Neural Operator”*. doi.org/10.5281/zenodo.7927198.
5. [Zenodo] **Peishi Jiang** et al. (Jan 2022 online) *Data and scripts associated with the manuscript “Using Mutual Information for Global Sensitivity Analysis on Watershed Modeling”*. doi.org/10.15485/1873128.
6. [Github] **Peishi Jiang** et al. (March 2021 online) *DART-PFLOTTRAN: An Ensemble-based Data Assimilation System for Estimating Subsurface Flow and Transport Model Parameters*. github.com/PeishiJiang/DART-PFLOTTRAN.
7. [Github] **Peishi Jiang** and Praveen Kumar (Aug 2019 online) *An Information-Theoretic Framework for Causal History Analysis in Complex System Dynamics*. github.com/PeishiJiang/info.
8. [Bitbucket] **Peishi Jiang** et al. (June 2017 online) *EMELI-Web: Experimental Modeling Environment for Linking and Interoperability - Web*. opensource.ncsa.illinois.edu/bitbucket/projects/ECGS/repos/emeli-web-application.

AWARDS

- | | |
|---|------------|
| • Bestie Award, Earth and Biological Sciences Directorate, PNNL | 2020, 2023 |
| • Award of Merit, NASA Frontier Development Laboratory Program | 2021 |
| • Entropy Travel Award, MDPI | 2018 |
| • CSDMS Student Scholarship | 2017 |